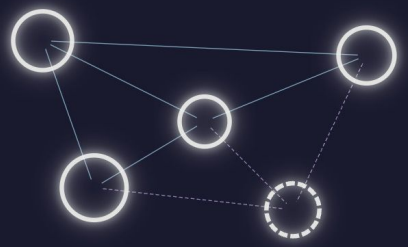


Thesis defense
Rob Romijnders

October 12th, 2026



Learning from Private Data:
Algorithms for Decentralized and Differentially Private Machine Learning

We live in a period of data-hungry AI

AI has rapid adoption among consumers

33% of general EU population used AI chatbots in 2025 (and 64% of 16-24 y.o.)

AI is trained on lots of data, and gets better with more data

Frontier models are trained on 30+ trillion tokens and see returns from more data

Yet the majority of data is private and behind firewalls

Studies estimate that 50% to 90% of data is private and unused

if only this data could be accessed without compromising privacy

https://ec.europa.eu/eurostat/web/products-eurostat-news/w/edn-20260210-1
Villalobos et al. ICML 2024; Kaplan et al. 2020
https://www.cycloneinteractive.com/sites/cyclone/assets/File/idc-digital-universe-infobrief.pdf
https://www.ibm.com/think/topics/ai-ready-data
http://www.ibmbigdatahub.com/blog/big-data-challenge-transformation-manufacturing-industry

if only this data could be accessed without compromising privacy

This thesis presents six studies towards this goal:

“Learning from Private Data”

(1) Decentralized contact tracing

(2) Contact tracing under differential privacy

(3) Neural augmentation differential privacy

(4) Differential privacy in 2-layer perceptrons

(5) Differential privacy in Mixture-of-Experts models

(6) Privacy in the use of Large Language models

Illustration

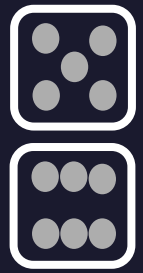
Privacy-sensitive questions:

- “Did you vote vote for this party in the election, yes or no?”
- “Is your salary higher than x, yes or no?”
- “Do you have medical condition x, yes or no?”
- “Do you like this product, yes or no?”

These are example yes/no questions and serve as illustration for bigger tasks such as real-valued scores, or neural network weights.

Noisy answer to yes/no question

Plausible deniability

1/3		Fixed to **“yes”**
1/3		Fixed to **“no”**
1/3		Tell the **local answer**

Noisy response to yes/no question

1/3		Fixed to "yes"
1/3		Fixed to "no"
1/3		Tell the local answer

126 out of 200 responded YES

What does that tell us?

Still, everyone has "plausible deniability"

Noisy response to yes/no question

1/3		Fixed to "yes"
1/3		Fixed to "no"
1/3		Tell the local answer

126 out of 200 responded YES

What does that tell us?

Estimate of 89%

and everyone has "plausible deniability"

Elements of a “noisy thing” in this thesis

Chapters 2, 3, and 4: the “noisy thing” is the risk score of COVID19

Chapters 5 and 6: the “noisy thing” are the weights of a neural network

Chapter 7: the “noisy thing” is the response from a Large Language Model

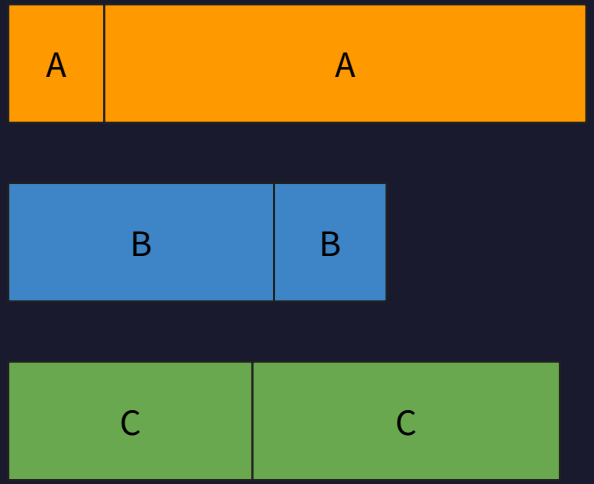
A case for decentralization

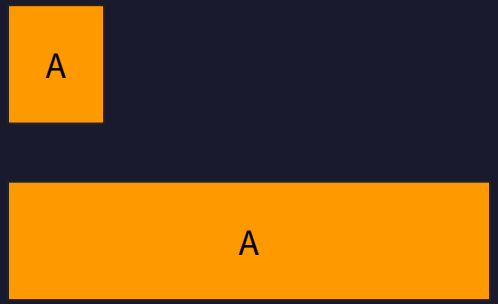
- Centralization of information is often not necessary
- Centralization carries risk of data breaches and hacks

Illustrated on the next slide with an example

“What is the average salary in this room?”

Central hub not required!







Step 1: send data among others

Step 2: sum with others

Step 3: average

“What is the average salary in this room?”

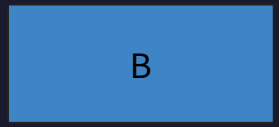
Central hub not required!







Step 1: send data among others







Step 2: sum with others



Step 3: average

“What is the average salary in this room?”

Central hub not required!

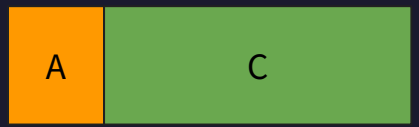






Step 1: send data among others







Step 2: sum with others



Step 3: average

Decentralized and privacy-preserving contact tracing

Economic impact of COVID 19 pandemic was enormous

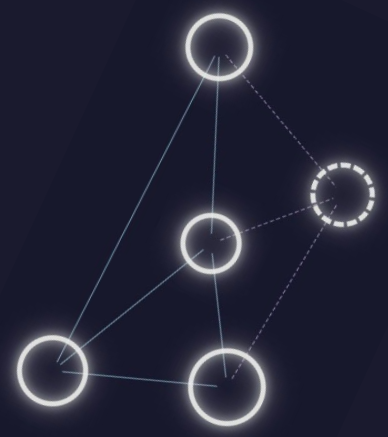
Studies estimate a trillion USD economic loss and mental-health consequences

The main cited reason for low adoption is privacy concerns

Among population studies in three countries totaling 7,000+ respondents

We propose a decentralized algorithm with DP guarantees

Study received oral presentation at AAAI-24 conference

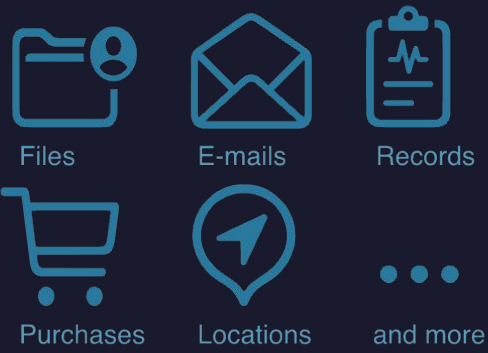


Overview

Beyond current open data,



there is untapped data to further improve AI systems:



This raises challenges

towards which we make the following contributions:

Trust & Safety

- Decentralized contact tracing — Ch. 2
  - ... with differential privacy — Ch. 3
  - ... and general purpose neural networks — Ch. 4

Efficiency

- Privately learning models without revealing intermediate steps — Ch. 5

Privacy in LLMs

- Modular LLM with private knowledge transfer — Ch. 6
- Private in-context learning for LLMs — Ch. 7

Supplementary slides

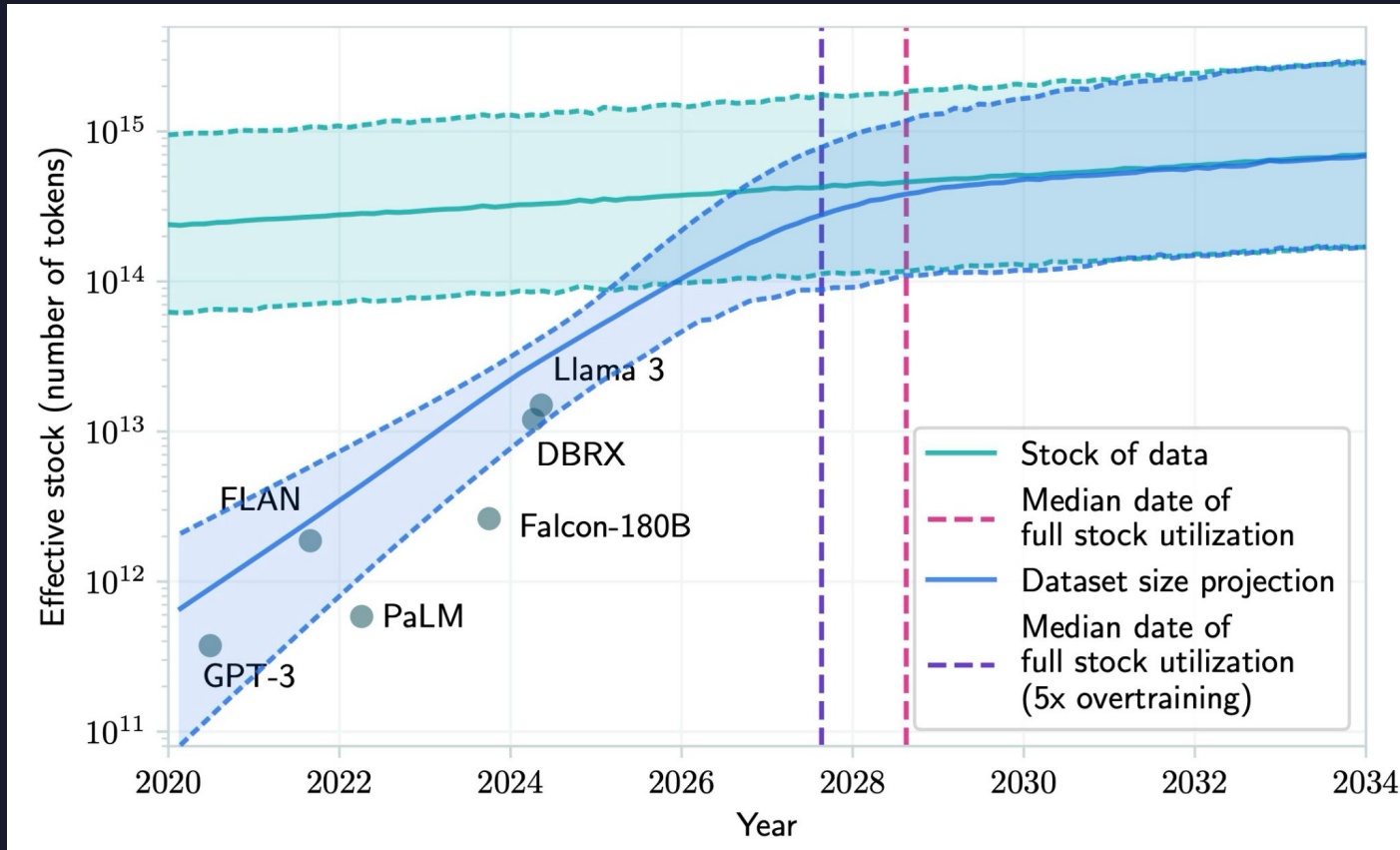
Slides after this are supplementary

Appendix slides

Training on web-scale data, what after?

Expectations are that frontier models will soon be training on all publicly available web data

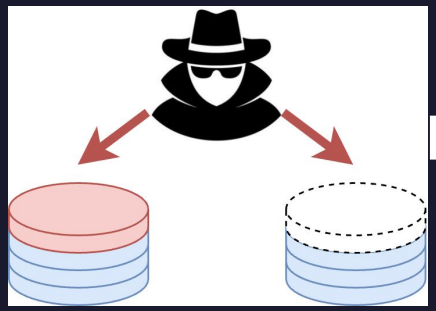
This indicates the need for methods to train on local and private sources of data



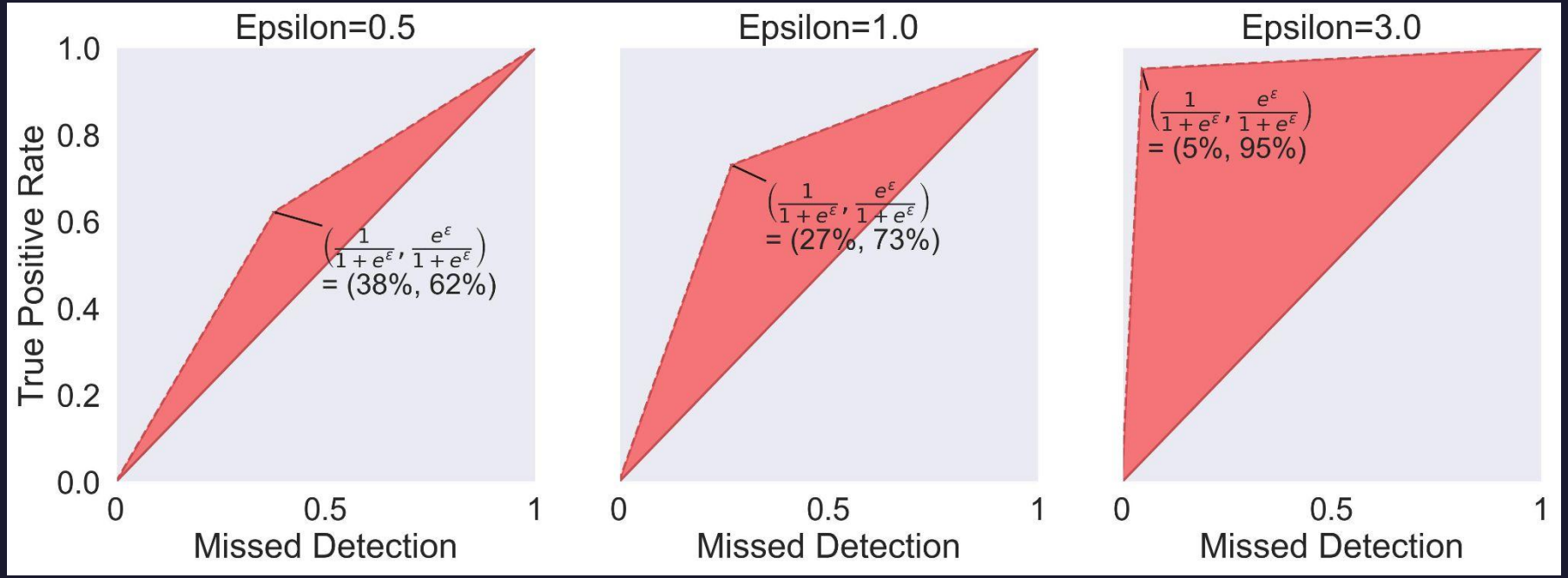








P mean?



https://en.wikipedia.org/wiki/Differential_privacy#Hypothesis_testing_interpretation